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Examination of 3D Self-Supervised Learning for Medical Imaging

Introduction

Medical image segmentation is difficult to scale because high-quality 3D labels require expert annotation, which is expensive and time-consuming. This is especially true for 3D MRI and CT data because each example is an entire volume instead of just one 2D image. Since medical datasets are often small, it is useful to have methods that can learn from unlabeled data. My project reproduces and adapts “3D Self-Supervised Methods for Medical Imaging” by Aiham Taleb et al. The paper studies whether self-supervised proxy tasks designed for 3D medical images can learn useful representations before the model is trained on a supervised medical task. The paper focuses on several proxy tasks, including Contrastive Predictive Coding (CPC), rotation prediction, jigsaw puzzles, relative patch location, and exemplar networks. The main contribution of the paper is showing that 3D self-supervised pretraining can improve performance, convergence speed, and label efficiency compared to training from scratch or using 2D slice-based pretraining. My project keeps the main idea of the paper, but I applied it to hippocampus segmentation and added a hybrid Rotation + CPC model. The reason I combined these two tasks is because rotation helps the model learn global 3D structure, while CPC helps the model learn local relationships between different parts of the volume.

Chosen Result

The result I focused on was the paper’s claim that 3D self-supervised pretraining can improve downstream segmentation performance compared with training from scratch. This result is important because the paper is not only saying that self-supervised tasks can be created in 3D. It is also saying that these tasks can make the encoder learn useful features before labels are used. For my project, I tested this idea on hippocampus segmentation from 3D MRI data. I compared a model using pretrained encoder weights against a model trained from scratch. I measured performance using Dice score, which is an overlap score between the predicted segmentation mask and the real ground-truth mask. A Dice score of 1 means the masks overlap perfectly, while a Dice score of 0 means there is no overlap. This result was chosen because it directly tests the main point of the paper: unlabeled 3D medical images contain useful structure that a model can learn before supervised training.

Methodology

My implementation followed a simplified version of the paper’s pipeline. First, I pretrained a 3D encoder using self-supervised learning. Then I transferred the encoder weights into a 3D U-Net-style segmentation model. After that, I fine-tuned the model on labeled hippocampus MRI data and compared the validation Dice score against a model trained from scratch. The 3D U-Net was used because it is designed for segmentation, where the encoder learns features from the input volume and the decoder turns those features into a voxel-level segmentation mask. My main change from the paper was using a hybrid Rotation + CPC pretraining setup. Rotation prediction works by rotating a 3D volume and making the model predict which rotation was applied. This is useful because the model has to learn the general shape and orientation of the volume. CPC works differently because it encourages the encoder to learn relationships between different feature regions and distinguish related regions from negative examples. I used a shared 3D encoder with two branches: one branch for rotation classification and one branch for a CPC-style contrastive objective. The combined loss was meant to help the encoder learn both orientation-based features and context-based features before being used for

segmentation. Since I had limited compute and storage, I used smaller subsets, fewer runs, and a reduced version of the full experimental setup from the original paper.

Results and Analysis

My pipeline was able to train and produce reasonable segmentation results, but I did not clearly reproduce the paper's strongest claim that self-supervised pretraining consistently beats training from scratch in low-label settings. The clearest trend was that performance improved when more labeled data was used, which makes sense because the model had more direct segmentation examples to learn from. The hybrid Rotation + CPC model showed that my self-supervised pretraining setup worked as a full pipeline, but the improvement over the scratch model was limited. The reason this likely happened is because self-supervised learning usually needs a lot of unlabeled data, longer training, and careful tuning before the pretrained features become very useful. CPC is also sensitive because the result depends on how positive and negative examples are chosen. Also, combining Rotation and CPC creates a balancing issue because one loss can dominate the other. If the model focuses too much on rotation, it may learn orientation shortcuts without learning useful local medical features. If the CPC part is too noisy, then it may not give a strong training signal. Overall, the hybrid model was a promising implementation, but its actual benefit was limited by compute, dataset size, training time, and the difficulty of matching the proxy tasks to hippocampus segmentation.

Reflections

This project showed me that reproducing medical imaging papers is difficult because the small implementation details matter a lot. The overall idea sounds simple: pretrain an encoder, transfer it into a segmentation model, and compare it against training from scratch. However, in practice, every step involved choices about preprocessing, model size, memory, proxy-task design, loss balancing, and training time. My main problem was the limited amount of data I had available. I also learned that 3D medical imaging is much more demanding than normal 2D computer vision because each input is a full volume, which makes training slower and more memory intensive. The hybrid Rotation + CPC model was useful because it let me test my own adaptation of the paper, but it also showed that combining self-supervised objectives does not automatically lead to better segmentation. Given more time and resources, I would compare Rotation-only, CPC-only, and hybrid pretraining directly, tune the hybrid loss more carefully, train on more unlabeled data, and run more seeds. Overall, my project successfully implemented the main workflow of self-supervised pretraining, encoder transfer, fine-tuning, and segmentation evaluation, even though it did not fully reproduce the strongest gains from the original paper.

References

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